

Appendix 1.1

An Elementary Discussion of Continuity, Differentiability, and Smoothness

Throughout Chapter 1 we have introduced mathematical terminologies and ideas in a gradual, as-needed format. Many of these ideas had to do with the continuity and differentiability of functions. The presentation was, admittedly, somewhat vague on these points in order that the main ideas would not be overencumbered. Careful characterization of the properties of functions is an essential ingredient in the development and analysis of finite element methods. However, to pursue this subject deeply would take us into the realm of serious mathematical analysis, which is outside the scope of this book. Nevertheless, we feel compelled to say a few additional words on the subject to round out the presentation in Chapter 1 and to expose the reader to notations and ideas that will probably be encountered if he or she attempts to read published papers on finite elements.

The discussion here will be restricted to one dimension. In Chapter 1 we spoke of continuously differentiable functions. If we have a grasp of the notion of a continuous function, then continuously differentiable functions pose no problem.

Definition: A function $f: \Omega \rightarrow \mathbb{R}$ (recall $\Omega =]0, 1[$) is said to be *k-times continuously differentiable*, or *of class* $C^k = C^k(\Omega)$, if its derivatives of order j , where $0 \leq j \leq k$, exist and are continuous functions.

A C^0 function is simply a continuous function. A C^∞ function is one that possesses a continuous derivative of any order (i.e., $j = 0, 1, \dots, \infty$).

Definition: A function f is said to be *of class* C_b^k if it is C^k and bounded (i.e., $|f(x)| < c$, where c is a constant, for all $x \in \Omega$).

Example 1

The functions defined by monomials (i.e., $f(x) = 1, x, x^2$, etc.) are C_r^r .

Example 2

The function $f(x) = 1/x$ is continuous on Ω , as are all its derivatives; hence it is C^∞ , but it is not bounded (i.e., there does not exist a constant c such that $|1/x| < c$ for all $x \in \Omega$; see Fig. 1.1.1). Consequently this function is not of class C_b^k for any $k \geq 0$.

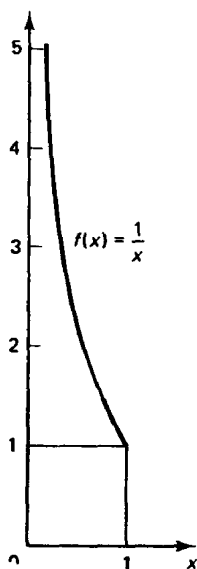


Figure 1.1.1 A continuous function that is not bounded.

Example 3

The function

$$f(x) = \begin{cases} x, & x \leq \frac{1}{2} \\ 1/2, & x > \frac{1}{2} \end{cases} \quad (1.1.1)$$

is continuous but not continuously differentiable (i.e., it is C_b^0 but not C_b^1). Functions in C_b^k , $k \geq 1$, but not in C_b^{k+1} may be constructed by integrating (1.1.1) k times. For example,

$$f(x) = \begin{cases} \frac{x^2}{2}, & x \leq \frac{1}{2} \\ \frac{(x - \frac{1}{4})}{2}, & x > \frac{1}{2} \end{cases} \quad (1.1.2)$$

is in C_b^1 but not C_b^2 . (The reader may wish to verify this.)

There is no universally accepted definition of what is meant by a “smooth” function. However, it is generally taken to mean that at least one derivative exists and is continuous (i.e., either C^1 or C_b^1) and sometimes means $k > 1$, even ∞ .

The C^k and C_b^k functions employ the classical notion of a derivative in their definitions. If we employ the closed unit interval, $\bar{\Omega} = [0, 1]$, instead of $\Omega =]0, 1[$, the difference between C^k and C_b^k disappears. This is because if f is $C^k([0, 1])$, $f(0)$ and $f(1)$ are real numbers and are not allowed to be ∞ . Thus unboundedness, as in the example above, is precluded. Very often, we think of C^k functions in this light. However, in some situations the differences between $C^k(\Omega)$ and $C_b^k(\Omega)$ must be kept in mind.

Generally, finite element functions are smooth on element interiors (there are exceptions, however) but possess only low-order continuity across element boundaries. One might be tempted to characterize them as locally smooth but globally "rough." The piecewise linear finite element functions discussed in Sec. 1.8 are of class C_b^0 . The Hermite cubics employed in Sec. 1.16 are C_b^1 . To calculate derivatives of such functions we need to employ the notion of a "generalized derivative," as was used in solving the Green's function problem of Sec. 1.10. For example, the first derivative of a piecewise linear finite element function is a generalized step function; the second derivative is a generalized Dirac delta function (i.e., delta functions, of various amplitudes, acting at the nodes). In the case of the Hermite cubics, the first derivative is continuous, the second a generalized step function, and so on. We have seen in Sec. 1.16 that other generalized functions also arise in the analysis of finite element behavior (namely, the dipole). The useful examples of generalized functions are by no means exhausted by what we have seen thus far. However, the ones we have introduced are perhaps the most basic.

In the mathematical analysis of boundary-value problems, and consequently in finite element analysis, we need to introduce classes of functions that possess generalized derivatives and, in addition, certain integrability properties. We have encountered such functions in the statements of weak formulations in Sec. 1.3 and 1.16. These are particular examples of *Sobolev spaces of functions* defined as follows:

$$H^k = H^k(\Omega) = \{w \mid w \in L_2; w_{,1} \in L_2; \dots; w_{,\underbrace{x_1 \dots x_k}_{k \text{ times}}} \in L_2\} \quad (1.1.3)$$

where

$$L_2 = L_2(\Omega) = \{w \mid \int_0^1 w^2 dx < \infty\} \quad (1.1.4)$$

In words, the Sobolev space of degree k , denoted by H^k , consists of functions that possess square-integrable generalized derivatives through order k . A square-integrable function is called an L_2 -function, by virtue of (1.1.4). From (1.1.3), we see that $H^0 = L_2$ and that $H^{k+1} \subset H^k$. The Sobolev spaces are the most important for studying elliptic boundary-value problems.

The question naturally arises as to the relation between Sobolev spaces and the classical spaces of differentiable functions introduced previously. In particular, when is an H^k -function smooth in the classical sense? The answer is provided by *Sobolev's theorem*, which states that, in one dimension, $H^{k+1} \subset C_b^k$. That is, if a function is of class H^{k+1} , then it is actually a C_b^k function. For example, in Sec. 1.3 we required H^1 functions. By Sobolev's theorem, such functions are, additionally, continuous and bounded. In Sec. 1.16, we employed H^2 functions. These are C_b^1 by Sobolev's theorem and thus possess bounded, continuous, classical derivatives.

Certain “singularities” are precluded by square-integrability. For example, $x^{-1/4}$ is in L_2 , but $x^{-1/2}$ is not. (Verify!) Such considerations become important in many physical circumstances (e.g., in fracture mechanics).

The number of other types of function spaces that arise in mathematical analysis is large, and many are difficult to comprehend without serious training in “functional analysis.” These topics are outside the scope of this book. The reader who wishes to delve further may consult [13, 14, 15] and references therein.

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